ETHZ Deep Learning 2026 Project Proposal
Project 14: How Much of the Learned Testing Policy Should a Model Keep?

Daniele Bugatti (`username`) [1] **Rafael van 't Veer (`username`)** [1] **Mehdi Ejtehadi (`username`)** [1]
Wesley Remy (`weremy`) [1]

Abstract

Clinical time-series models learn from *when* and *whether* tests are ordered, not only from their values. This testing pattern reflects one hospital's observation policy and changes when the model is deployed elsewhere. Before deployment, hospitals typically run a model silently for 1–3 months, which yields only a handful of labelled outcome cases. We propose a ladder of adaptations for a latent state-space model with a separate testing head: first re-learning that head from the new hospital's records, which needs no outcome labels, then the testing and outcome heads, then the full model. We ask which observation shifts can be corrected without labels, and how many outcome cases the others need.

1. Introduction & Related Work

Problem. Deep models for clinical time series learn from what is measured and also from when it is measured: a lactate order signals that a doctor suspects sepsis. This testing pattern is predictive (Sisk et al., 2021), but it reflects the hospital's observation policy, and it changes when a model trained at hospital A is deployed at hospital B (*observation shift*). The change comes in two kinds. Hospital B may test differently given what the model can see: troponin is ordered for 3% of patients at Beth Israel but 32% at Emory (PhysioNet 2019, Reyna et al., 2020), so an order means less there. Or B's clinicians may test on information the model never sees, so an order carries outcome information that B's records alone cannot reveal.

Before going live, models are often run silently for 1–3 months (Sendak et al., 2020). That run-in yields the only labelled data from B, and very little of it: a typical ICU admits 600–1,300 patients a year (Checkley et al., 2014), so a 1–3 month run-in contains about 2–6 sepsis cases. Outcomes are scarce, but B's records are not: they show which tests B orders for which patients. We therefore propose a **ladder of adaptations** for a latent state-space model in which the testing evidence enters through its own head: (1) re-learning the testing head from B's records without outcomes, the rest frozen; (2) re-learning the testing and outcome heads with the run-in's outcomes; (3) fine-tuning the whole model, starting from (1) and penalised towards it. Higher rungs repair more kinds of shift but need more outcome cases; rung 1 needs none.

Research question. *Which observation shifts can be corrected without outcome labels by re-learning only the testing part, and how many outcome cases do the higher rungs need for the rest?* Concretely: (i) does rung 1 beat the fixed rules of prior work (always keep or always remove the pattern) and full fine-tuning at realistic run-in sizes; (ii) for which kinds of shift, and from how many outcome events, do rungs 2 and 3 pay off; (iii) does the state-space model adapt better than GRU-D, whose testing part cannot be separated, fine-tuned at its last layer or in full.

Deep models use the observation pattern but do not separate it. Architectures for irregular series consume masks and time gaps (GRU-D, Che et al., 2018), continuous-time paths (Neural CDE, Kidger et al., 2020; Latent ODE, Rubanova et al., 2019) or time-value sets with attention (mTAN, Shukla & Marlin, 2021; Raindrop, Zhang et al., 2022), and fuse the pattern with physiology in one representation. Models with a separate observation component exist, a Hawkes process for clinicians' measurement events (Alaa et al., 2017) and DeepJoint's observation heads (Jeanselme et al., 2022), but none adapts that component alone at a new site.

Observation shift between hospitals. Multi-site deep sepsis and ICU benchmarks lose 0.08–0.09 AUROC across sites (Moor et al., 2023; van de Water et al., 2024) and include missingness indicators without attributing the loss to them. Studies that isolate the pattern choose all or nothing: Futoma et al. (2021) fitted with measurement indicators and dropped them at deployment, which transferred best from Beth Israel

Advisor TA: Sarah Martinson. Topic: Project 14, Deep Time-Series Models under Observation Shift. [1]Department of Computer Science, ETH Zurich, Zurich, Switzerland. Correspondence to: Wesley Remy <weremy@student.ethz.ch>.

Project for the Deep Learning 2026 Class.

to a Memphis hospital but not in reverse; Rockenschaub et al. (2024) found that ignoring informative missingness costs at the source but is more robust elsewhere, especially when missingness is highly informative. Lasko et al. (2024) argue conceptually for a site-specific process model next to a stable disease model; Zhou et al. (2023) adapt to missingness shift without labels in linear models. None re-learns the observation part of a deep sequence model at a new site.

Adapting with few labels. Clinical model updating is a ladder from recalibration to full refitting, and parsimonious updates win at small samples (Steyerberg et al., 2004; Vergouwe et al., 2017); a full refit is risky well below the 100 events needed just to validate a model (Riley et al., 2024). In deep learning, which part to fine-tune depends on the kind of shift (Lee et al., 2023), and retraining only the last layer can suffice (Kirichenko et al., 2023), but clinical fine-tuning studies use hundreds to thousands of target patients (Moor et al., 2023; van de Water et al., 2024). **Gap:** no work re-learns only the observation part of a clinical sequence model from unlabelled target records, and none compares such adaptations across run-in sizes and kinds of shift.

2. Experimental Setup & Methods

Model. A neural latent state-space model, extending our pilot: a discrete latent patient state ($K \approx 32$ states) evolves by a learned transition, and three heads read it: a value head (likelihood of each measured value), a testing head (probability that each test is ordered, given the state) and an outcome head. Predictions come from filtering the state over time, trained end to end on the outcome loss plus the likelihood of values and tests. The testing pattern enters only through the testing head. Rung 1 re-learns the testing head by maximising the likelihood of B's values and test orders, with initial distribution, transition, value and outcome heads frozen; rung 2 also re-learns the outcome head with the run-in's outcomes; rung 3 fine-tunes everything. Baselines: keep and remove the testing evidence (weight $w=1$ and $w=0$, Futoma), a single tuned weight w (our pilot), recalibration, and GRU-D, unchanged or fine-tuned at the last layer or in full.
E1: simulation with known optimum. Our latent-state simulator, whose exact Bayes predictor is known, with three shift types: B tests more or less selectively or at random (visible in B's records); B's clinicians also test on a hidden risk factor (visible only through outcomes); and the same patient state carries a different risk at B. We measure each rung's regret against the Bayes predictor at B over 1–104 weeks of deployment (up to about 50 events). Expected: rung 1 suffices for the first type; the others need rungs 2–3 and enough events.

E2: real hospitals (dataset to be determined). Requirements: several hospitals with hospital identifiers, time-stamped labs and vitals with the testing pattern preserved, and enough outcome events per site; the main candidate is eICU-CRD (208 US hospitals, Pollard et al., 2018; credentialed access). Design: ordered source $\rightarrow$ target pairs, run-ins of 1, 3, 6, 12 and 24 months drawn from the target site (20 draws each), evaluation on all remaining target patients, primary metric AUROC (pre-specified), secondary calibration.

E3: architecture. Whether the state-space model adapts from fewer cases than GRU-D fine-tuned at the last layer or in full, on the pairs of E2.

Computational cost and feasibility. We have one NVIDIA GTX 1080 Ti (11 GB) on snowflake and a 4-core laptop. All models are small (32 latent states or GRU hidden size 32–64, at most 48 hourly steps per patient), so GPU memory is not a constraint. Measured on the laptop CPU: the pilot state-space model trains in 15 minutes on 8,000 simulated patients, and 108 simulation fits of GRU, GRU-D, Neural CDE and Transformer took 3.6 hours in total; E1 therefore runs on the laptop. E2 and E3 need about 50 source models (sites $\times$ models $\times$ 3 seeds) and a few thousand short adaptation runs, which we run on the GPU. We will time one site first and, if the full design does not fit the available weeks, reduce the number of draws or pairs.

3. Results & Discussion

Preliminary results (linear models on open PhysioNet 2019 and 2012 data; simulation). On Beth Israel $\rightarrow$ Emory, keeping the pattern was best, because there was barely any observation shift between the two (pattern-model transfer gap 0.03 AUROC, 95% CI -0.01 to 0.07); dropping it cost 0.05 AUROC. Across 26 unit pairs the pattern's meaning shifted clearly in 7, mostly for mortality; there, a single weight tuned on a 3–24 month run-in gained up to 0.02–0.03 AUROC and stayed within 0.001 elsewhere (after removing impossible lab values). A one-parameter update beat a full refit up to about 20 cases; the refit won from about 50. In simulation, rung 1 from 25 unlabelled patients came close to the exact optimum under four testing shifts (regret 0.0001–0.0003 Brier, keep up to 0.0012), and full fine-tuning was worse up to two years; with a hidden risk factor, only fine-tuning helped, after two years.

4. Conclusion

The project turns the keep-or-remove question of prior work into re-learning the testing part from the new hospital's own records, with outcome-based adaptation only where the run-in supports it. The simulator, the pilot model and the run-in pipeline already exist. Timeline: weeks 1–2 the multivariate state-space model (with static covariates) and E1; weeks 3–4 E2 and E3; weeks 5–6 analysis and report.

References

Alaa, A. M., Hu, S., and van der Schaar, M. Learning from clinical judgments: Semi-Markov-modulated marked Hawkes processes for risk prognosis. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pp. 60–69, 2017.

Che, Z., Purushotham, S., Cho, K., Sontag, D., and Liu, Y. Recurrent neural networks for multivariate time series with missing values. *Scientific Reports*, 8:6085, 2018. doi: 10.1038/s41598-018-24271-9.

Checkley, W., Martin, G. S., Brown, S. M., et al. Structure, process, and annual ICU mortality across 69 centers: United States Critical Illness and Injury Trials Group Critical Illness Outcomes Study. *Critical Care Medicine*, 42(2):344–356, 2014. doi: 10.1097/CCM.0b013e3182a275d7.

Futoma, J., Simons, M., Doshi-Velez, F., and Kamaleswaran, R. Generalization in clinical prediction models: The blessing and curse of measurement indicator variables. *Critical Care Explorations*, 3(7):e0453, 2021. doi: 10.1097/CCE.0000000000000453.

Jeanselme, V., Martin, G., Peek, N., Sperrin, M., Tom, B., and Barrett, J. DeepJoint: Robust survival modelling under clinical presence shift. NeurIPS Workshop on Learning from Time Series for Health, 2022. arXiv:2205.13481.

Kidger, P., Morrill, J., Foster, J., and Lyons, T. Neural controlled differential equations for irregular time series. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

Kirichenko, P., Izmailov, P., and Wilson, A. G. Last layer re-training is sufficient for robustness to spurious correlations. In *International Conference on Learning Representations*, 2023.

Lasko, T. A., Strobl, E. V., and Stead, W. W. Why do probabilistic clinical models fail to transport between sites. *npj Digital Medicine*, 7:53, 2024. doi: 10.1038/s41746-024-01037-4.

Lee, Y., Chen, A. S., Tajwar, F., Kumar, A., Yao, H., Liang, P., and Finn, C. Surgical fine-tuning improves adaptation to distribution shifts. In *International Conference on Learning Representations*, 2023.

Moor, M., Bennett, N., Plečko, D., Horn, M., Rieck, B., Meinshausen, N., Bühlmann, P., and Borgwardt, K. Predicting sepsis using deep learning across international sites: a retrospective development and validation study. *eClinicalMedicine*, 62:102124, 2023. doi: 10.1016/j.eclinm.2023.102124.

Pollard, T. J., Johnson, A. E. W., Raffa, J. D., Celi, L. A., Mark, R. G., and Badawi, O. The eICU collaborative research database, a freely available multi-center database for critical care research. *Scientific Data*, 5:180178, 2018. doi: 10.1038/sdata.2018.178.

Reyna, M. A., Josef, C. S., Jeter, R., Shashikumar, S. P., Westover, M. B., Sharma, A., Nemati, S., and Clifford, G. D. Early prediction of sepsis from clinical data: the PhysioNet/Computing in Cardiology Challenge 2019. *Critical Care Medicine*, 48(2):210–217, 2020. doi: 10.1097/CCM.0000000000004145.

Riley, R. D., Snell, K. I. E., Archer, L., et al. Evaluation of clinical prediction models (part 3): calculating the sample size required for an external validation study. *BMJ*, 384: e074821, 2024. doi: 10.1136/bmj-2023-074821.

Rockenschaub, P., Xian, Z., Zamanian, A., et al. Robust prediction under missingness shifts, 2024. arXiv:2406.16484.

Rubanova, Y., Chen, R. T. Q., and Duvenaud, D. Latent ordinary differential equations for irregularly-sampled time series. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Sendak, M. P., Ratliff, W., Sarro, D., et al. Real-world integration of a sepsis deep learning technology into routine clinical care: implementation study. *JMIR Medical Informatics*, 8(7):e15182, 2020. doi: 10.2196/15182.

Shukla, S. N. and Marlin, B. Multi-time attention networks for irregularly sampled time series. In *International Conference on Learning Representations*, 2021.

Sisk, R., Lin, L., Sperrin, M., Barrett, J. K., Tom, B., Diaz-Ordaz, K., Peek, N., and Martin, G. P. Informative presence and observation in routine health data: A review of methodology for clinical risk prediction. *Journal of the American Medical Informatics Association*, 28(1): 155–166, 2021. doi: 10.1093/jamia/ocaa242.

Steyerberg, E. W., Borsboom, G. J. J. M., van Houwelingen, H. C., Eijkemans, M. J. C., and Habbema, J. D. F. Validation and updating of predictive logistic regression models: a study on sample size and shrinkage. *Statistics in Medicine*, 23(16):2567–2586, 2004. doi: 10.1002/sim.1844.

van de Water, R., Schmidt, H., Elbers, P., Thoral, P., Arnrich, B., and Rockenschaub, P. Yet another ICU benchmark: A flexible multi-center framework for clinical ML. In *International Conference on Learning Representations*, 2024.

Vergouwe, Y., Nieboer, D., Oostenbrink, R., Debray, T. P. A., Murray, G. D., Kattan, M. W., Koffijberg, H., Moons, K. G. M., and Steyerberg, E. W. A closed testing procedure to select an appropriate method for updating prediction models. *Statistics in Medicine*, 36(28):4529–4539, 2017. doi: 10.1002/sim.7179.

Zhang, X., Zeman, M., Tsiligkaridis, T., and Zitnik, M. Graph-guided network for irregularly sampled multivariate time series. In *International Conference on Learning Representations*, 2022.

Zhou, H., Balakrishnan, S., and Lipton, Z. Domain adaptation under missingness shift. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 9577–9606, 2023.